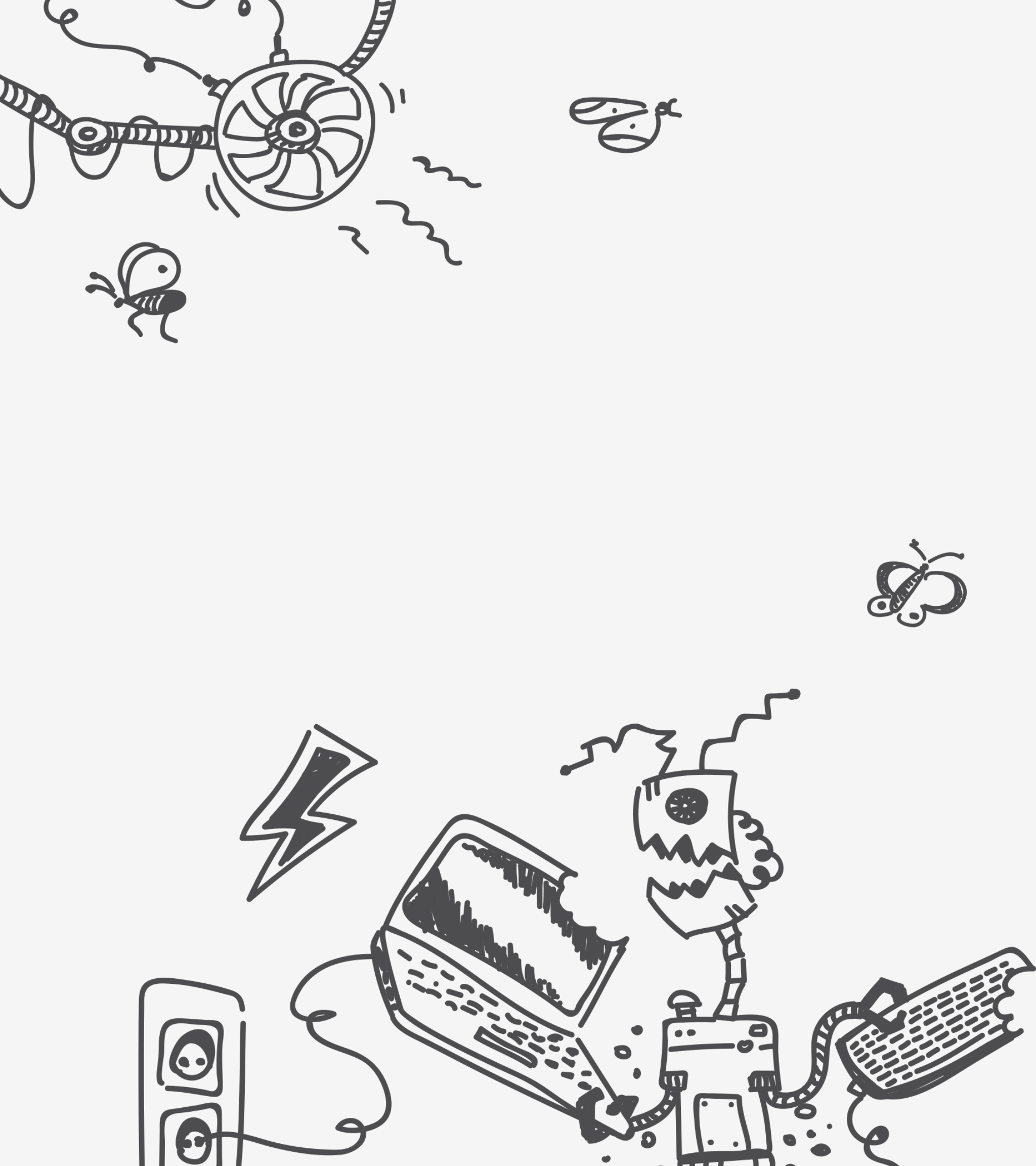
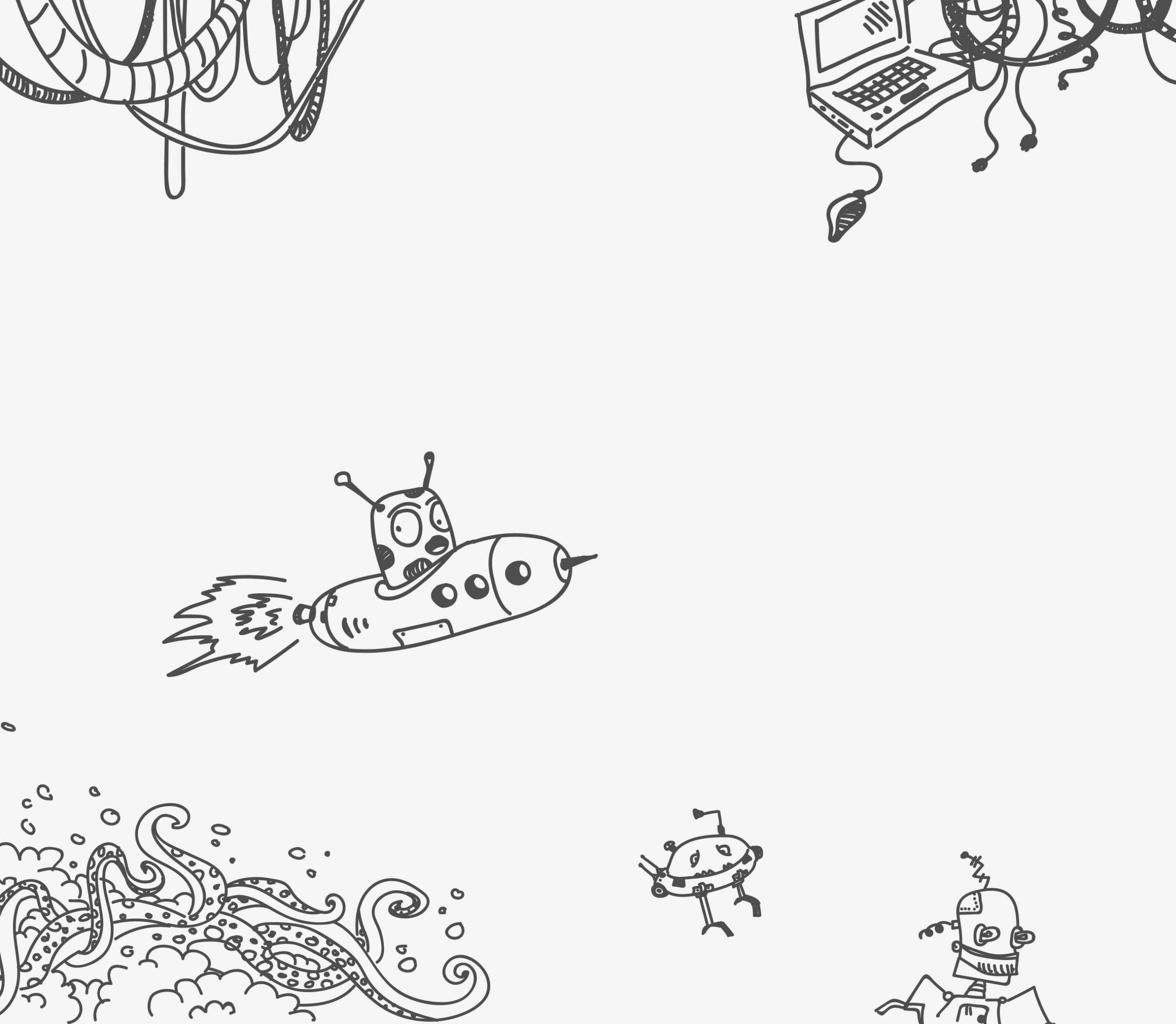




Decision tree in regression - a theoretical introduction

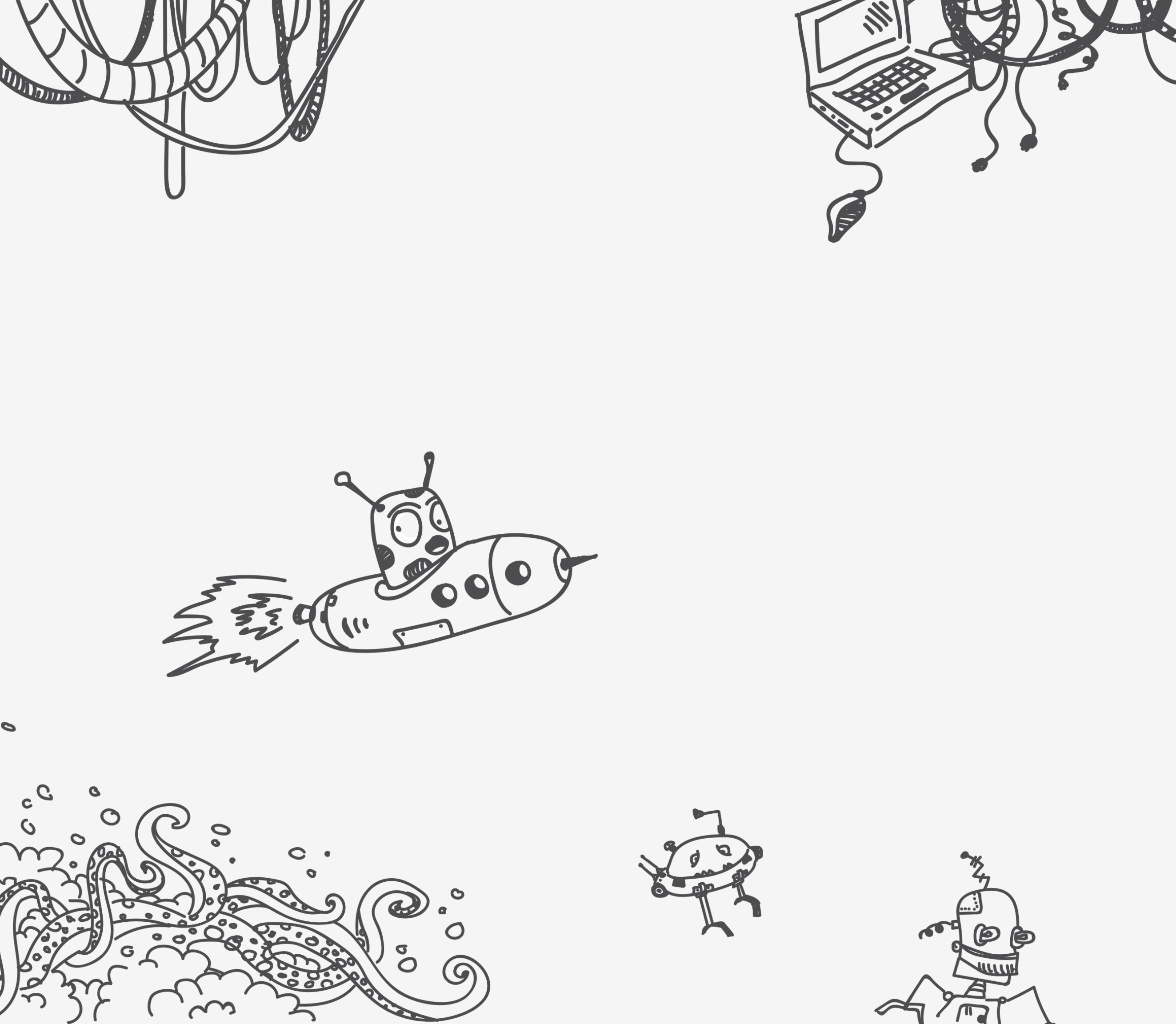


Introduction

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We already have a good chunk of knowledge about the basic algorithms that are used to model the regression problem. Before we finish the first day of the **Machine Learning** course, we need to discuss one more algorithm: the decision tree. The decision tree is one of the most widely used approaches in supervised learning. It can be used for both regression and classification problems. The way of solving problems using a decision tree is quite different compared to linear or polynomial regression.

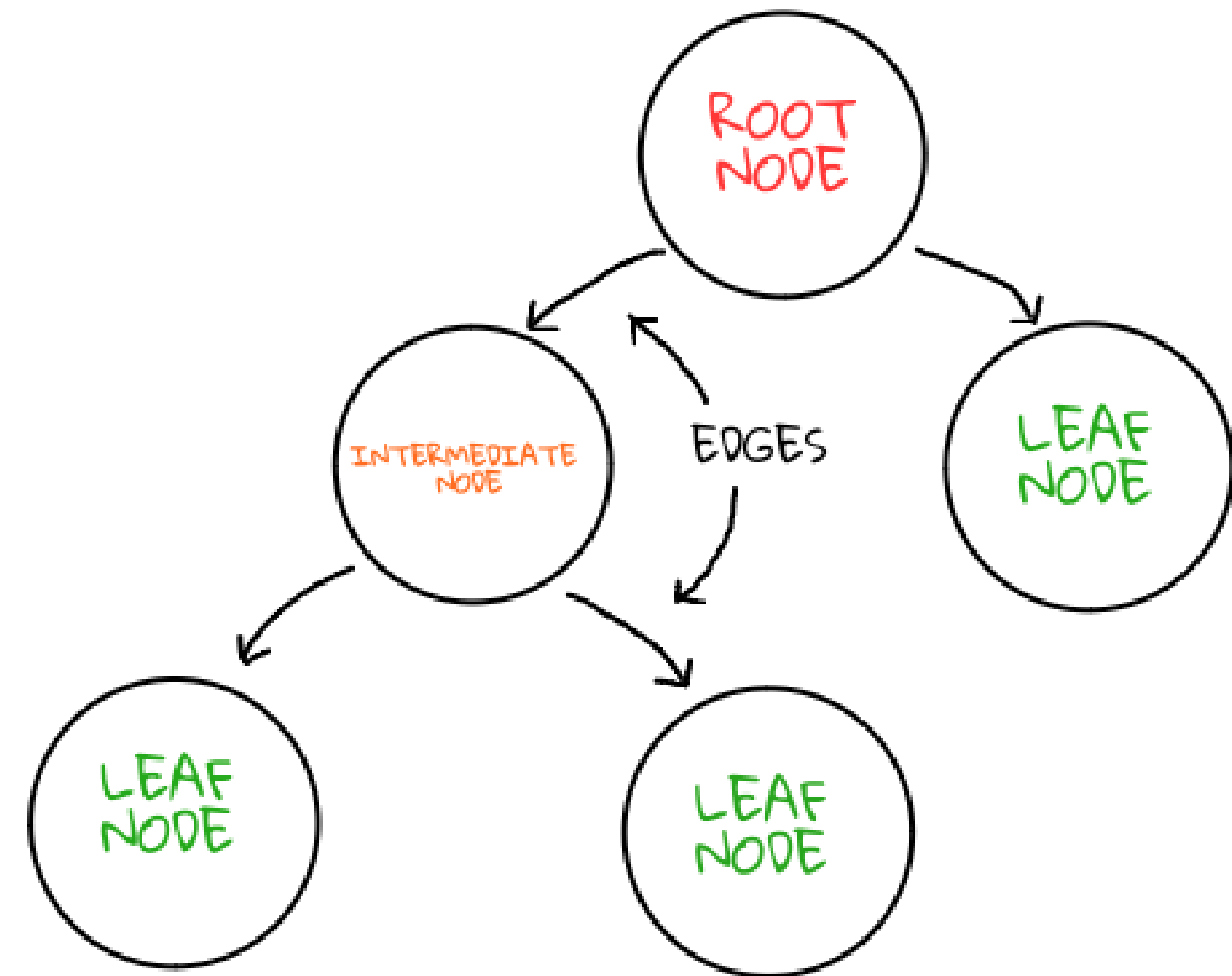
Let's get down to specifics and start by explaining how a decision tree works.



**Where did the name
"decision tree" come
from?**

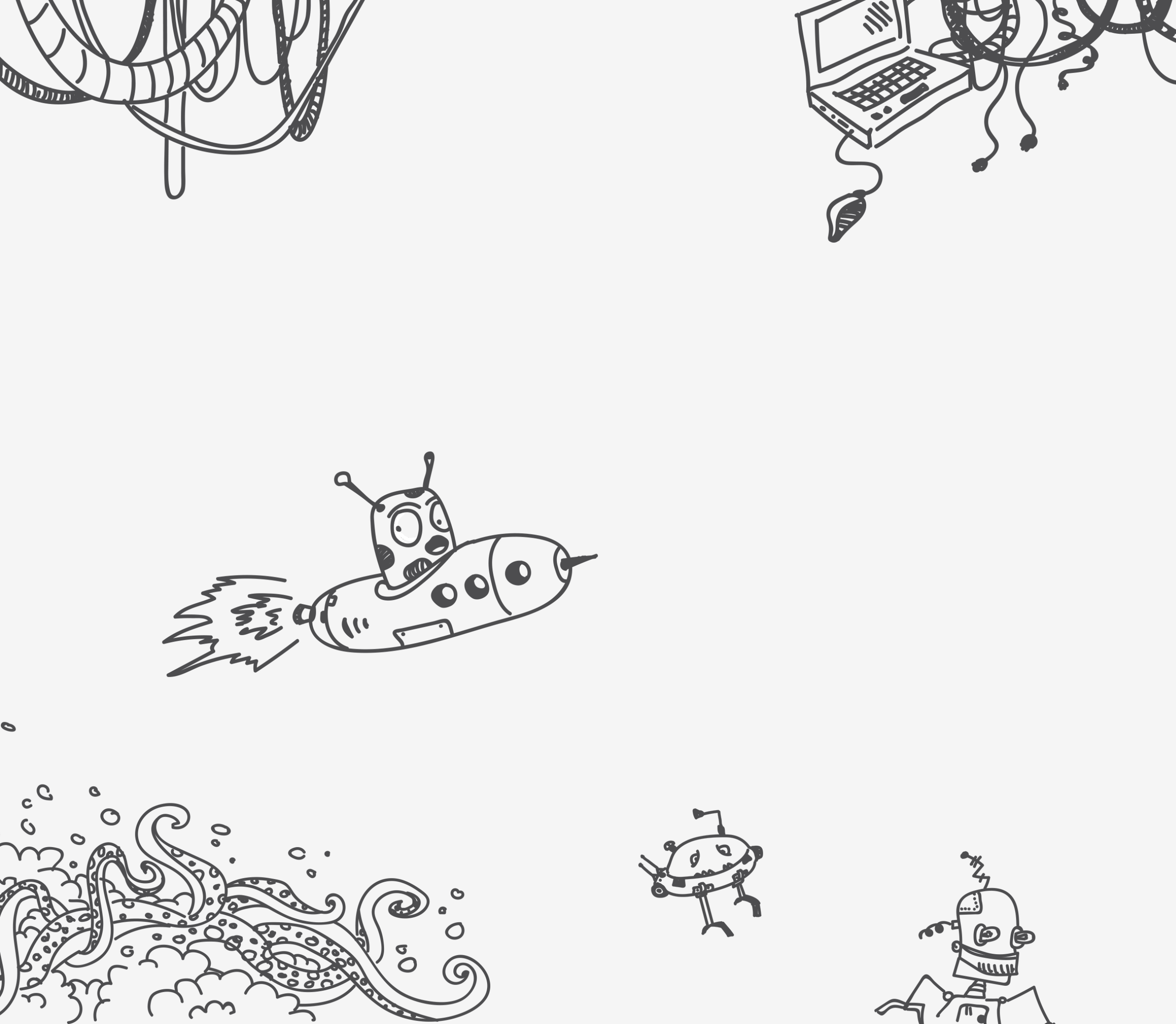
Where did the name "decision tree" come from?

The term *decision tree* was used to name the algorithm we are discussing because of its structure. We can visualize the algorithm as a tree, consisting of nodes and edges (branches) as elements that connect each node.



In the image, we can distinguish three types of nodes:

- **root node** - this is the initial node from which the process of splitting the data begins. Here the data is divided on the basis of the variable, which from the point of view of the algorithm is the most relevant



How is a decision tree trained?

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A decision tree is built by dividing our training data in the best possible recursive way, using the values of each variable. On what basis do we conclude that a particular partitioning is best at a particular location? This is done by calculating metrics at each split. For decision trees and regressions, the default metric is mean squared error (**MSE**), but we can also use mean absolute error (**MAE**). Each split is implemented in such a way as to minimize the mean error value.

The decision tree handles both categorical variables (in general, discrete variables) and continuous variables. In the case of categorical variables (e.g., the color of a car), the model uses the values of the variables explicitly as attributes on which it makes divisions.

For continuous variables, the decision tree makes the split by comparing the variable's value with a certain threshold. If its value is less than or equal to the *threshold*, the object goes into one group, and if the value is greater - into another. An example of such a variable can be the mileage of a car.

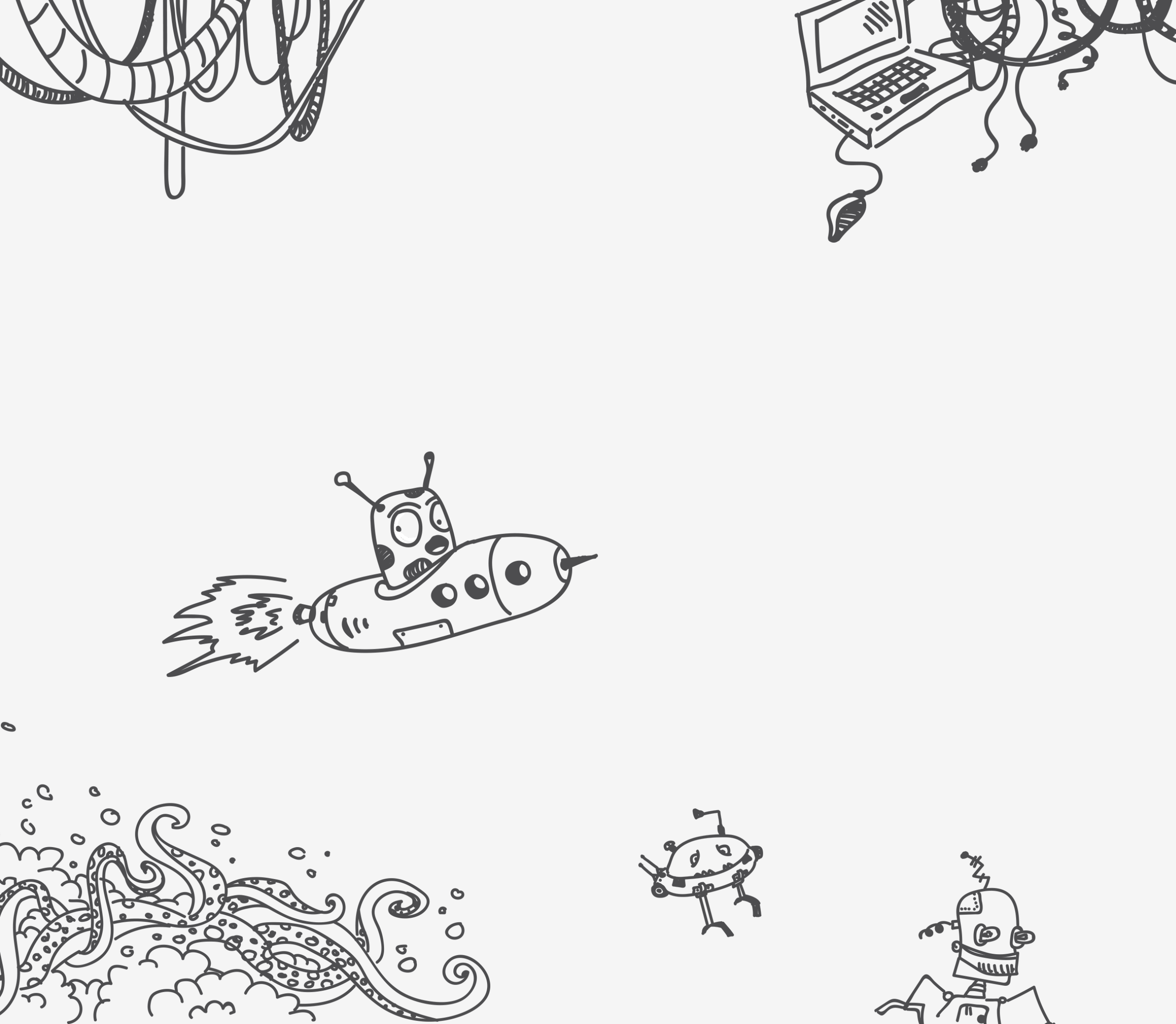
How is a decision tree trained?

When training a decision tree, the selection of hyperparameters of the model is very important. Hyperparameters in the decision tree model include:

- the depth of the tree (i.e., the number of splits),
- The minimum number of observations at a node,
- the splitting criterion (e.g., **MSE/MAE** for regression),
- the number of variables considered for the model.

The selection of appropriate hyperparameter values has a major impact on the quality of the prediction. Trees that make too many splits, include many variables, or require a small number of observations at the nodes, tend to overfit. For this reason, when building models based on a decision tree, we want to pruning (pruning) the tree by setting a limit on the maximum depth of the tree.

On the other hand, trees with a very simple structure, make the model too simple and cannot generalize well the complex relationships in the data.

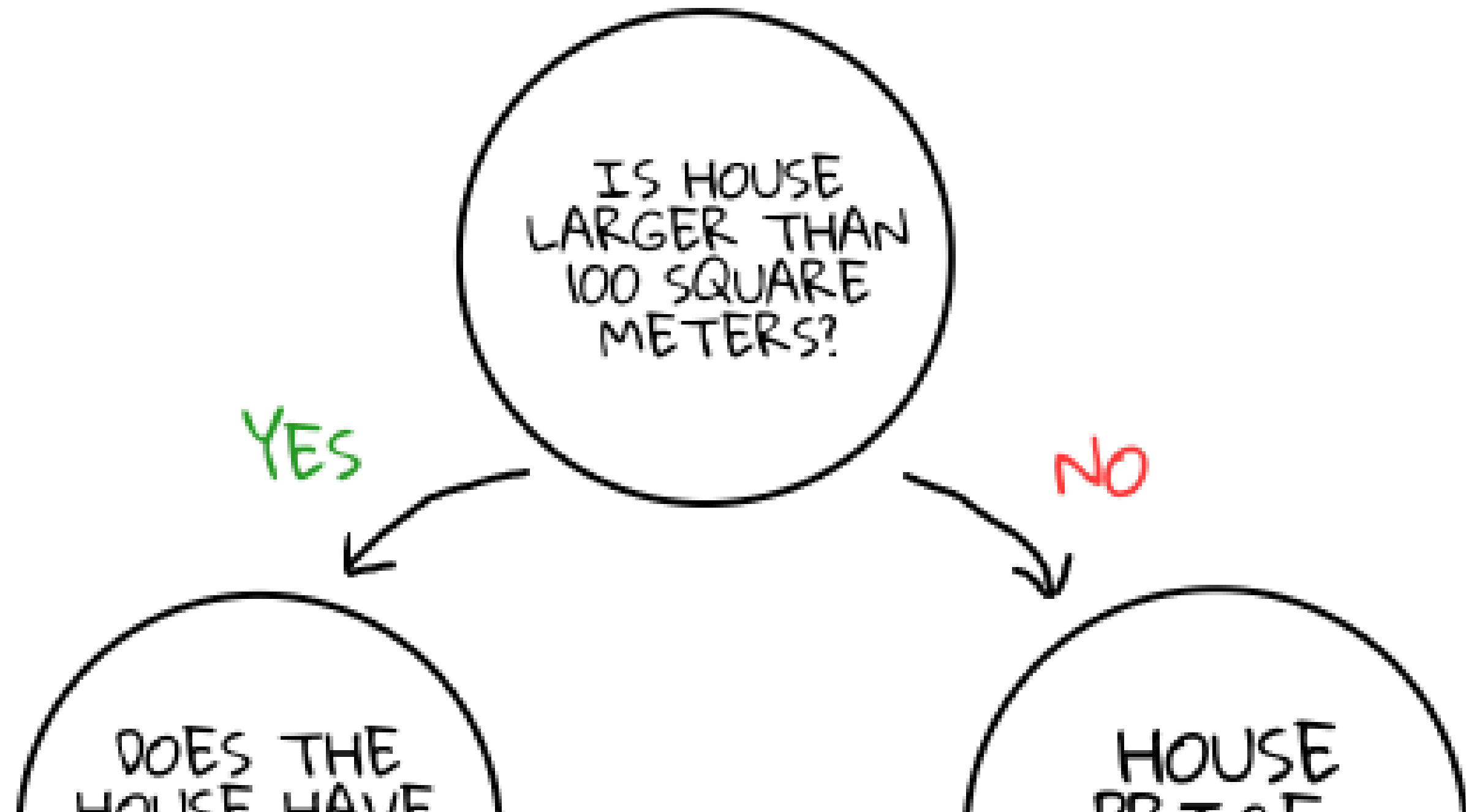


Decision tree by example

Decision tree by example

When creating a decision tree, at each node of the tree we ask a different question about the variable we have chosen to split. Based on the question we ask, we calculate the answer, which can be an intermediate answer that will be used for the next division of the data or the final answer.

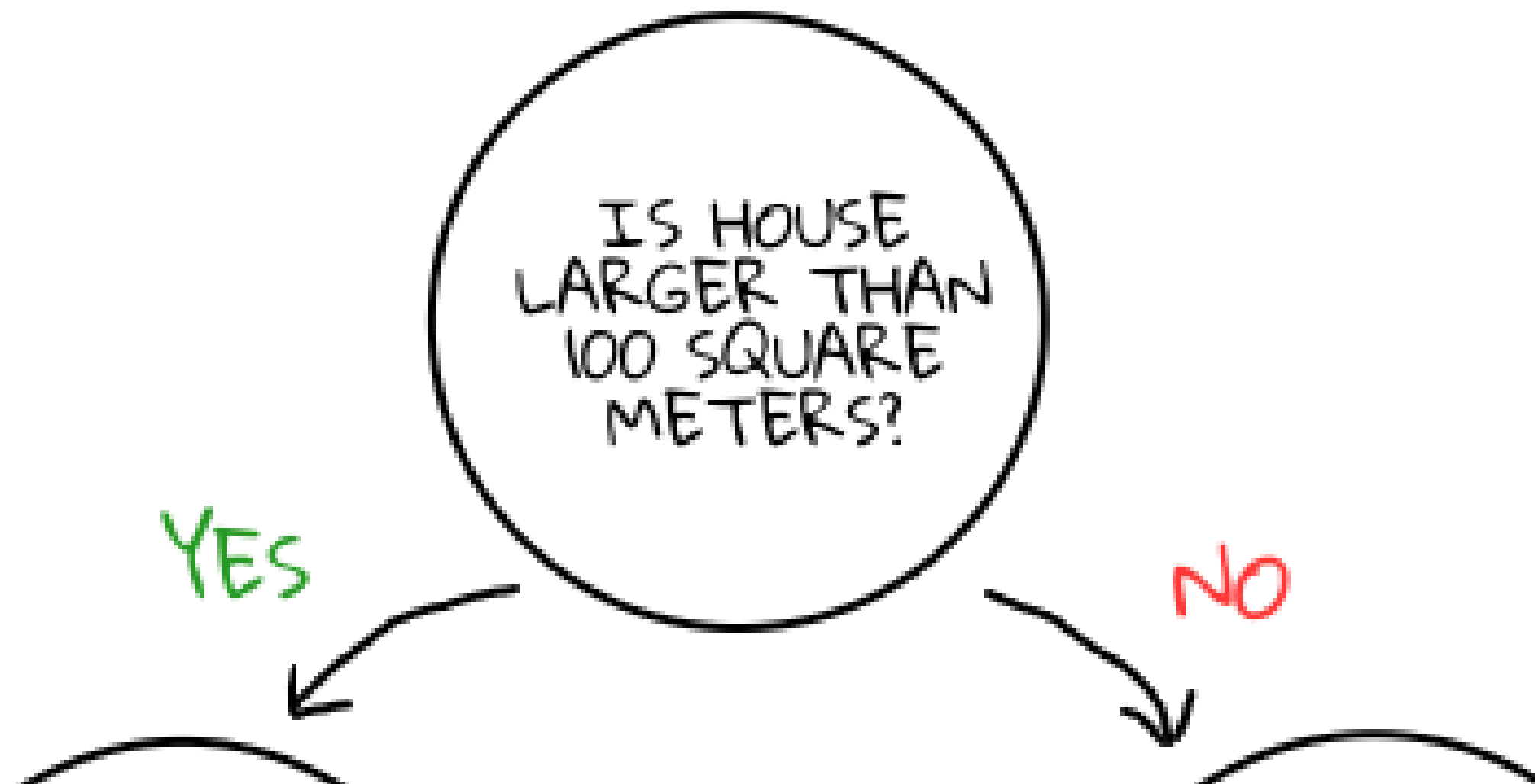
Let's try to understand this in the following example. Suppose we want to predict the price of a house based on its parameters - area or number of rooms.



Decision tree by example

We first split our house data by area. Why first by area and not by number of bedrooms? That's because the model has determined that it is the more important variable in terms of influencing the final outcome, the price of the property. The model's choice coincides with reality. Sellers often quote the price of a property per square meter. Surely you have never seen the price quoted in terms of the number of rooms.

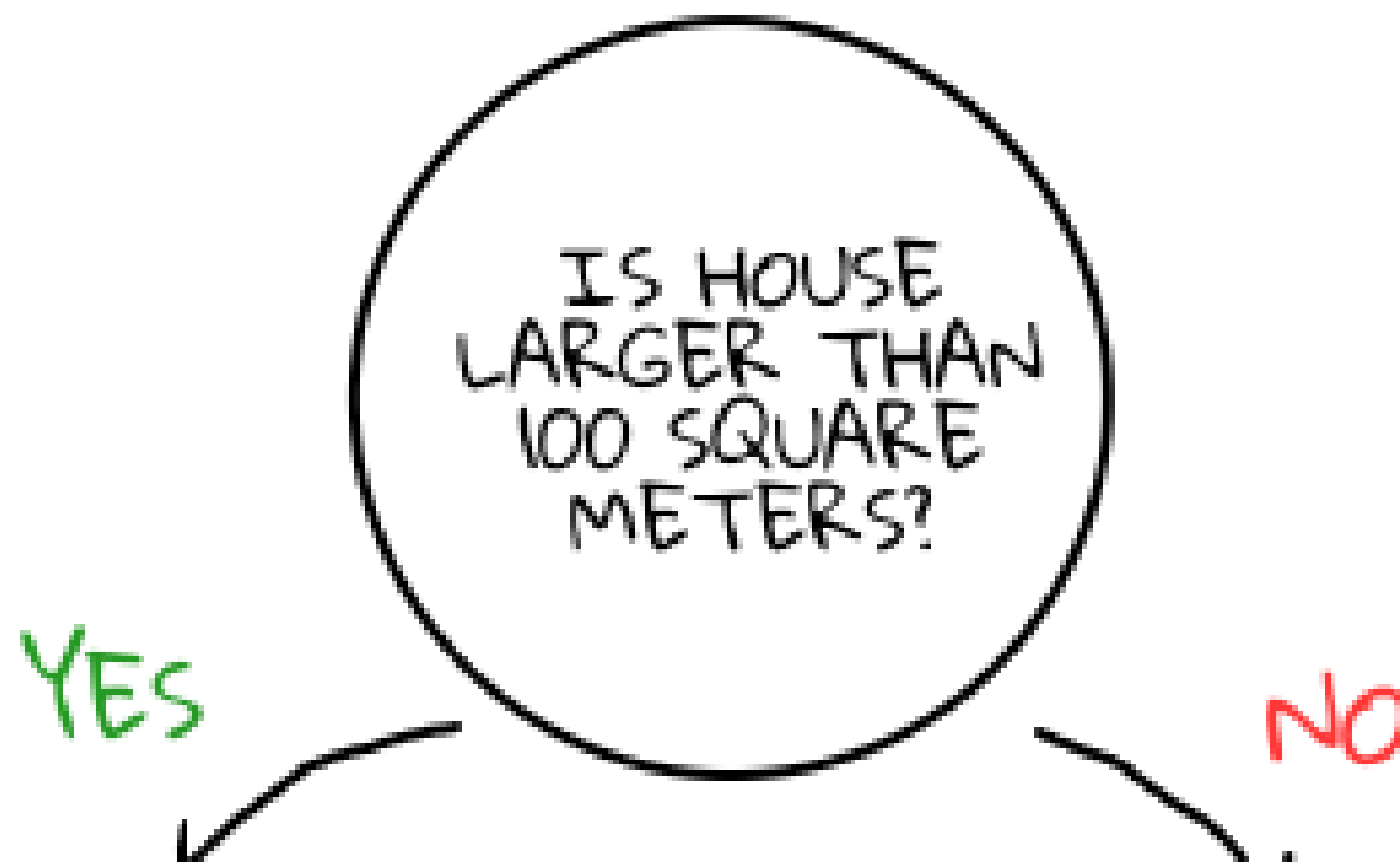
Based on the area, the model makes a split - it took 100 square meters as the area threshold. If the house is more than 100 square meters, it further searches for the price of the house, based on the next division. If the house is smaller, the model predicts that the price of the house is \$400,000.

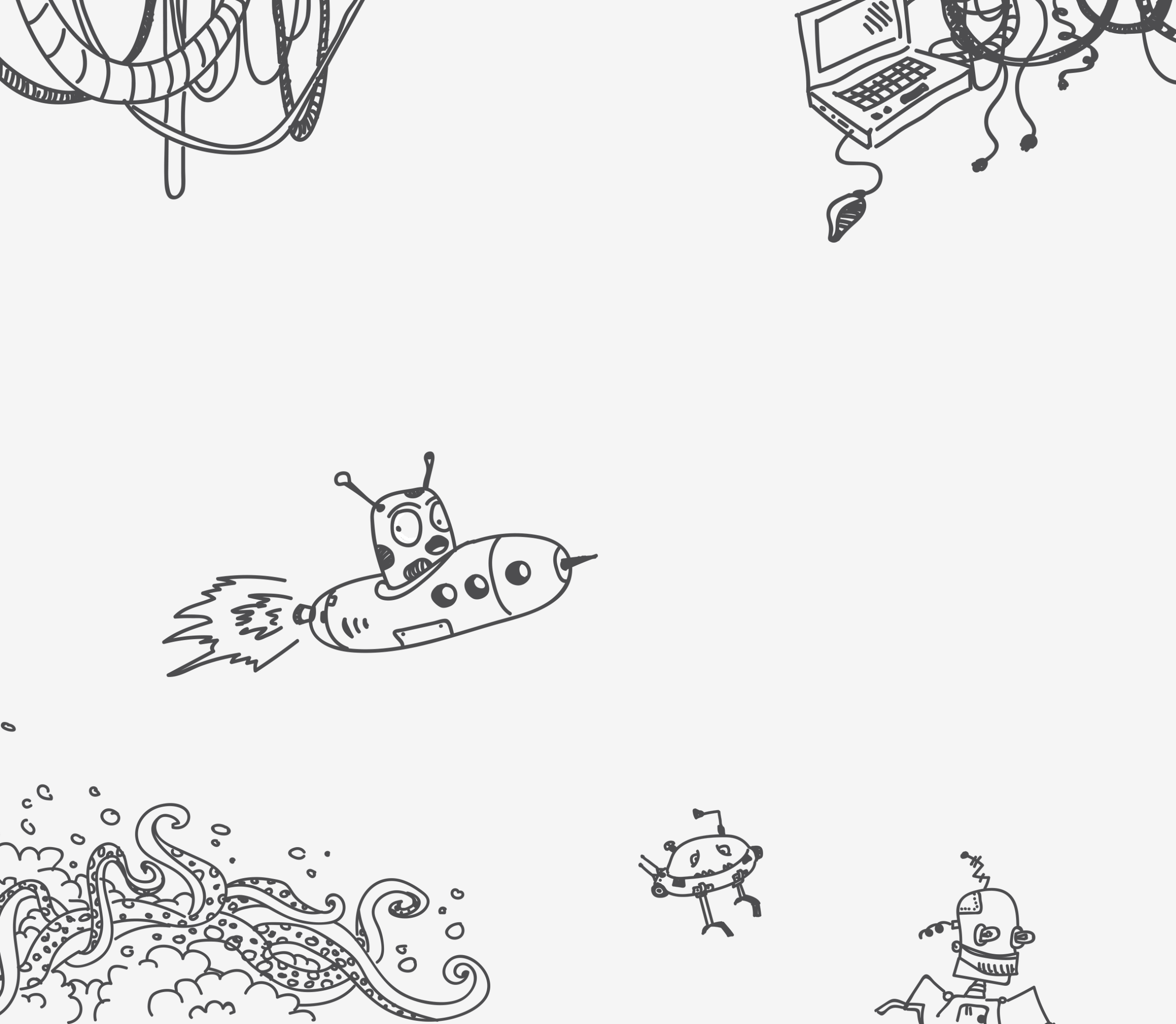


Decision tree by example

Now let's look at the next split, which takes into account houses larger than 100 square meters. The model again selects a threshold, this time for the number of bedrooms. If the house has more than 3 bedrooms, the price is 500,000, *but if the house has 3 bedrooms (or less), the price is 450,000*. The values, found at the leaf nodes, are the average value of all observations at that node. Assuming that in the training data we had five houses of more than 100 sqm and more than three bedrooms with prices in that order (in dollars): 500 000, 600 000, 400 000, 620 000, 380 000.

The above example is, of course, simplified, because in reality the price of houses is affected by many more variables. However, we used an uncomplicated case to foster an understanding of how the decision tree model calculates the result.





**What are the
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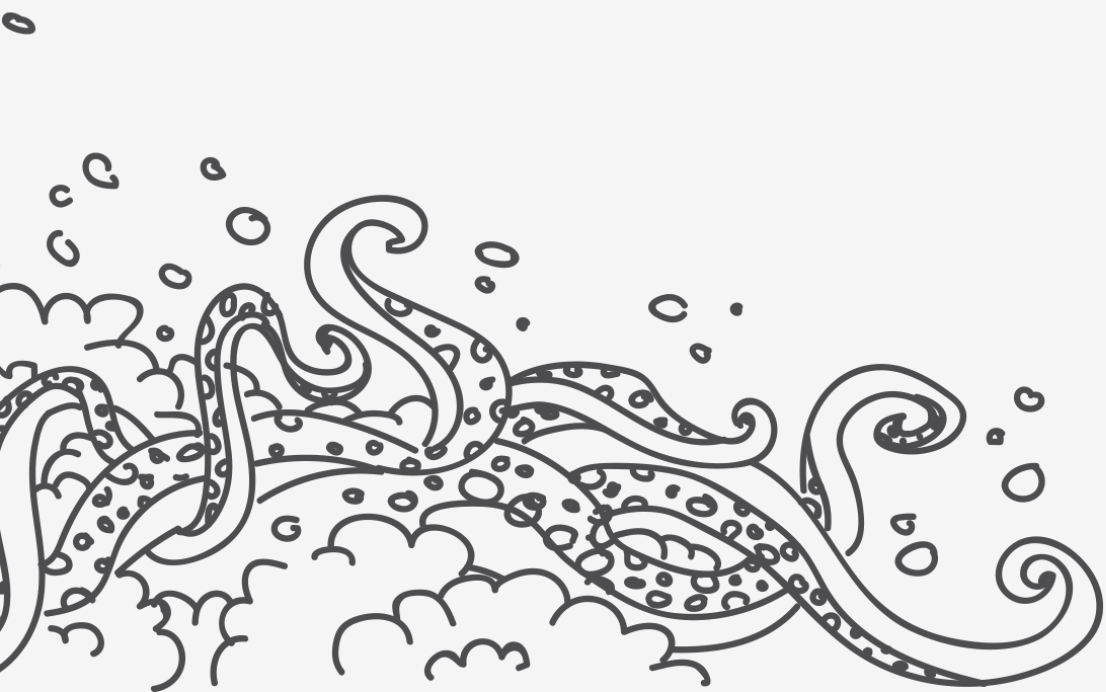
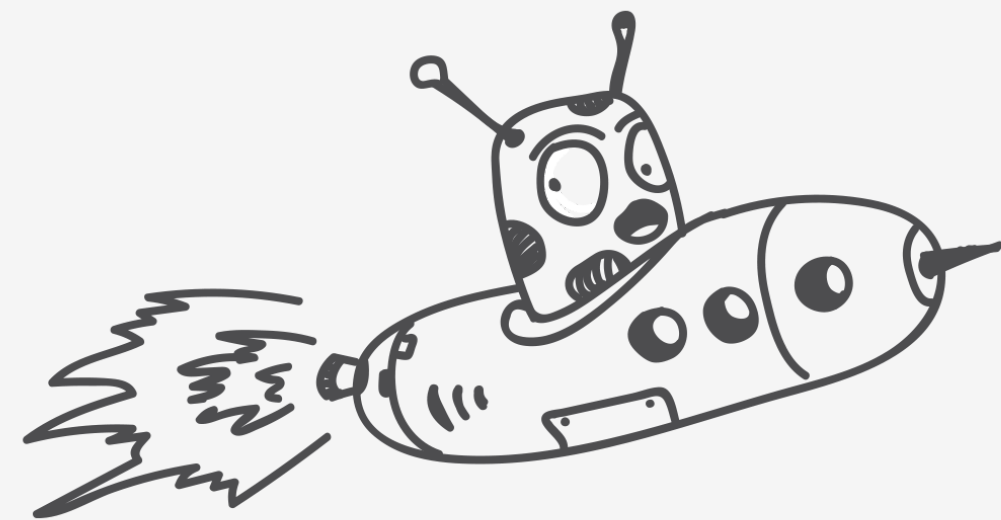
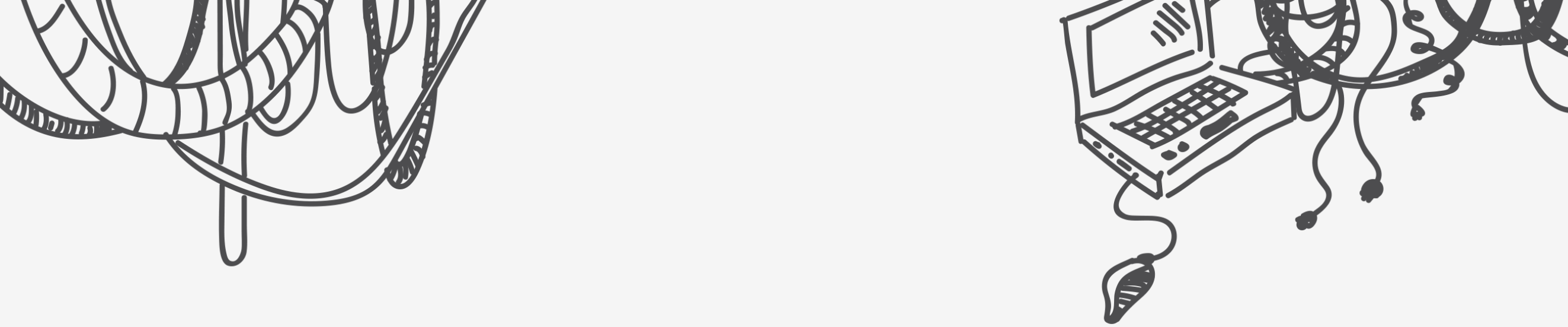
Nothing is perfect, and it is exactly the same with a decision tree. So let's summarize the advantages and disadvantages of using this algorithm.

ADVANTAGES

- Relatively simple algorithm to understand. This makes it easy to interpret the decisions it has made. The way decisions are made (making data splits) can be visualized with a tree. We can also easily get information about the impact of individual variables on the result of the model.
- It does not require as time-consuming preparation of data for training as with other algorithms. The decision tree can handle missing values or categorical variables. The use of dummy encoding, for example, can negatively affect the model's predictions.
- It can be used for both regression and classification, with much more frequent use in solving the latter problem.

DISADVANTAGES

- A decision tree is a more computationally complex algorithm than, for example, linear or polynomial regression.
- Trees with low complexity may have poor generalization ability.
- It may have a tendency to overfit the training data. For this reason, it is necessary to be careful with the selection of hyperparameters such as the depth of the tree or the minimum number of observations used for partitioning.
- They are sometimes unstable, i.e. the addition of new observations can cause a complete reconstruction of the tree.



Summary

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We presented the last of the algorithms discussed on the first day of the Machine Learning course. The decision tree is an easy-to-understand algorithm used in both regression and classification. The training data for the decision tree does not require preparation as meticulous as the other supervised learning methods discussed earlier.

Missing and outlier values or encoding of categorical variables is not a problem here (the latter is even inadvisable, due to the rapidly increasing number of splits). A decision tree is computationally more complex than, for example, linear regression, and tends to overtrain, so proper selection of hyperparameters when training the model is critical.